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WEEK NO. 5-

PART A: EMBEDDINGS FUNDAMENTALS

Exercises:-

1. Compare Similar and Unrelated Meaning

Code:

```
embed_demo.py > ...  
72     sentence_1 = (  
73         "A young wizard attends a magic school"  
74     )  
75  
76     sentence_2 = (  
77         "A boy learns spells at an academy"  
78     )  
79  
80     sentence_3 = (  
81         "A recipe for chocolate cake"  
82     )  
83  
84  
85     v1 = embed(sentence_1)  
86     v2 = embed(sentence_2)  
87     v3 = embed(sentence_3)  
88  
89  
90     similar_score = cosine_similarity(  
91         v1,  
92         v2  
93     )  
94  
95     unrelated_score = cosine_similarity(  
96         v1,  
97         v3  
98     )
```

```

embed_demo.py > ...
111     print("\n--- Similarity Results ---")
112
113     print(
114         "Similar meaning:",
115         similar_score
116     )
117
118     print(
119         "Unrelated meaning:",
120         unrelated_score
121     )

```

Output:

```

--- Similarity Results ---
Similar meaning: 0.6272689100537832
Unrelated meaning: 0.015505222027939176

```

2. Inspect the Embedding Vector

Code:

```

embed_demo.py > ...
124     print("\n--- Embedding Information ---")
125
126     print(
127         "Length of v1:",
128         len(v1)
129     )
130
131     print(
132         "First 10 values of v1:"
133     )
134
135     print(
136         v1[:10]
137     )

```

Output:

```

(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartA-Embeddings-Fundamentals\embedding-demo> .\.venv\Scripts\python.exe embed_demo.py
Sentence 1:
A young wizard attends a magic school

Sentence 2:
A boy learns spells at an academy

Sentence 3:
A recipe for chocolate cake

--- Similarity Results ---
Similar meaning: 0.6272689100537832
Unrelated meaning: 0.015505222027939176

--- Embedding Information ---
Length of v1: 384
First 10 values of v1:
[0.022908316925168037, 0.05676201358437538, -0.04974302276968956, -0.022635124623775482, -0.024459393694996834, -0.013513431884348392, 0.028800517320632935, -0.016749419271945953, -0.0351775623857975, -0.043942298740148544]
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartA-Embeddings-Fundamentals\embedding-demo>

```

3. Similar Words but Opposite Meaning

Code:

```

opposite_meaning_test.py > ...
71 sentence_positive = (
72 |     "I loved this book"
73 | )
74
75 sentence_negative = (
76 |     "I did not love this book"
77 | )
78
79
80 positive_vector = embed(
81 |     sentence_positive
82 | )
83
84 negative_vector = embed(
85 |     sentence_negative
86 | )
87
88
89 score = cosine_similarity(
90 |     positive_vector,
91 |     negative_vector
92 | )
93
94
95 print("Sentence 1:")
96 print(sentence_positive)
97
98 print("\nSentence 2:")
99 print(sentence_negative)
100
101 print(
102 |     "\nCosine similarity:",
103 |     score
104 | )

```

Output:

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Wexa Solutions\Week 5\PartA-Embeddings-Fundamentals\embedding-demo> .\.venv\Scripts\python.exe opposite_me
aning_test.py
Sentence 1:
I loved this book

Sentence 2:
I did not love this book

Cosine similarity: 0.782282845690016
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Wexa Solutions\Week 5\PartA-Embeddings-Fundamentals\embedding-demo>
```

PART B: VECTOR DATABASES

Exercises:-

1. Confirm Space Query Returns Science Fiction First

Code:

```
chroma_demo.py > ...
12 collection.add(
13     documents=[
14         "A young wizard attends a magic school and fights a dark lord.",
15         "A crew travels through a wormhole to save humanity from a dying Earth.",
16         "Two feuding families in 19th century England navigate love and marriage.",
17     ],
18     metadatas=[
19         {
20             "source": "book_1.txt",
21             "category": "Fantasy"
22         },
23         {
24             "source": "book_2.txt",
25             "category": "Science Fiction"
26         },
27         {
28             "source": "book_3.txt",
29             "category": "Romance"
30         },
31     ],
32     ids=[
33         "book_1",
34         "book_2",
35         "book_3"
36     ],
37 )
38
39
40 results = collection.query(
41     query_texts=[
42         "a space journey story"
43     ],
44     n_results=2
45 )
```

```

chroma_demo.py > ...
48  print(
49      "\n--- Vector Search Results ---"
50  )
51
52  print(
53      "Documents:"
54  )
55
56  print(
57      results["documents"]
58  )
59
60  print(
61      "\nMetadata:"
62  )
63
64  print(
65      results["metadatas"]
66  )

```

Output:

```

(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartB-Vector-Databases\chroma-demo> .\venv\Scripts\python.exe chroma_demo.py
--- Vector Search Results ---
Documents:
[['A crew travels through a wormhole to save humanity from a dying Earth.', 'Two feuding families in 19th century England navigate love and marriage.']]
Metadata:
[[{'source': 'book_2.txt', 'category': 'Science Fiction'}, {'source': 'book_3.txt', 'category': 'Romance'}]]
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartB-Vector-Databases\chroma-demo>

```

2. Metadata Filtering in Chroma

Code:

```

metadata_filter_demo.py > ...

40 results = collection.query(
41     query_texts=[
42         "magic adventure"
43     ],
44     n_results=1,
45     where={
46         "category": "Fantasy"
47     }
48 )
49
50
51 print(
52     "\n--- Filtered Search Results ---"
53 )
54
55 print(
56     "Documents:"
57 )
58
59 print(
60     results["documents"]
61 )
62
63 print(
64     "\nMetadata:"
65 )
66
67 print(
68     results["metadatas"]
69 )

```

Output:

```

(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Wexa Solutions\Week 5\PartB-Vector-Databases\chroma-demo> .\venv\Scripts\python.exe metadata_filter_demo.py

--- Filtered Search Results ---
Documents:
[['A young wizard attends a magic school and fights a dark lord.']]

Metadata:
[[{'category': 'Fantasy', 'source': 'book_1.txt'}]]
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Wexa Solutions\Week 5\PartB-Vector-Databases\chroma-demo>

```

3. Adding More Documents and Repeating Vector Searches

Code:

expanded_chroma_demo.py > ...

```
12 collection.add(  
13     documents=[  
14         "A young wizard attends a magic school and fights a dark lord.",  
15         "A crew travels through a wormhole to save humanity from a dying Earth.",  
16         "Two feuding families in 19th century England navigate love and marriage.",  
17         "A detective investigates a mysterious murder in a quiet English village.",  
18         "A group of explorers discover ancient ruins hidden deep inside a jungle.",  
19     ],  
20     metadatas=[  
21         {  
22             "source": "book_1.txt",  
23             "category": "Fantasy"  
24         },  
25         {  
26             "source": "book_2.txt",  
27             "category": "Science Fiction"  
28         },  
29         {  
30             "source": "book_3.txt",  
31             "category": "Romance"  
32         },  
33         {  
34             "source": "book_4.txt",  
35             "category": "Mystery"  
36         },  
37         {  
38             "source": "book_5.txt",  
39             "category": "Adventure"  
40         },  
41     ],  
42 ],  
43 )
```

expanded_chroma_demo.py > ...

```
56 queries = [  
57     "a murder investigation",  
58     "an adventure exploring ancient places",  
59     "a story about travelling through space"  
60 ]  
61  
62  
63 for query in queries:  
64  
65     print(  
66         "\n===== "  
67     )  
68  
69     print(  
70         "Query:",  
71         query  
72     )  
73  
74  
75     results = collection.query(  
76         query_texts=[  
77             query  
78         ],  
79         n_results=1  
80     )  
81  
82  
83     print(  
84         "Best match:"  
85     )  
86  
87     print(  
88         results["documents"][0][0]  
89     )
```

expanded_chroma_demo.py > ...

```
92     print(  
93         "Metadata:"  
94     )  
95  
96     print(  
97         results["metadatas"][0][0]  
98     )
```

Output:

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartB-Vector-Databases\chroma-demo> .\.venv\Scripts\python.exe expanded_chroma_demo.py

=====
Query: a murder investigation
Best match:
A detective investigates a mysterious murder in a quiet English village.
Metadata:
{'source': 'book_4.txt', 'category': 'Mystery'}

=====
Query: an adventure exploring ancient places
Best match:
A group of explorers discover ancient ruins hidden deep inside a jungle.
Metadata:
{'category': 'Adventure', 'source': 'book_5.txt'}

=====
Query: a story about travelling through space
Best match:
A crew travels through a wormhole to save humanity from a dying Earth.
Metadata:
{'category': 'Science Fiction', 'source': 'book_2.txt'}
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartB-Vector-Databases\chroma-demo>
```

PART C: RAG FUNDAMENTALS: BUILDING THE PIPELINE BY HAND

Exercises:-

1. Add 3–4 Documents Using add_document()

Code:

```
rag_pipeline.py > ...
55 def chunk_text(
56     text: str,
57     chunk_size: int = 300,
58     overlap: int = 50
59 ) -> list[str]:
60
61     chunks = []
62
63     start = 0
64
65     while start < len(text):
66
67         end = start + chunk_size
68
69         chunk = text[start:end]
70
71         chunks.append(
72             chunk
73         )
74
75         start += (
76             chunk_size - overlap
77         )
78
79     return chunks
```

```
rag_pipeline.py > ...
105 def add_document(
106     text: str,
107     source: str
108 ):
109
110     chunks = chunk_text(
111         text
112     )
113
114     embeddings = [
115         embed(chunk)
116         for chunk in chunks
117     ]
118
119     collection.add(
120         documents=chunks,
121
122         embeddings=embeddings,
123
124         metadatas=[
125             {
126                 "source": source,
127                 "chunk": index
128             }
129             for index in range(
130                 len(chunks)
131             )
132         ],
133
134         ids=[
135             f"{source}-{index}"
136             for index in range(
137                 len(chunks)
138             )
139         ]
140     )
```

```

rag_pipeline.py > ...
239 def load_documents():
240
241     documents_folder = Path(
242         "documents"
243     )
244
245     for file_path in (
246         documents_folder
247         .glob("*.txt")
248     ):
249
250         text = file_path.read_text(
251             encoding="utf-8"
252         )
253
254         add_document(
255             text=text,
256             source=file_path.name
257         )
258
259         print(
260             f"Added document: {file_path.name}"
261         )

```

Output:

```

(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartC-Manual-RAG-Pipeline\rag-demo> .\venv\Scripts\python.exe rag_pipeline.py
Added document: magic_school.txt
Added document: mars_mission.txt
Added document: ocean.txt
Added document: rainforest.txt

```

2. Ask an Answerable Question

Code:

rag_pipeline.py > ...

```
143 def retrieve(  
144     question: str,  
145     k: int = 3  
146 ):  
147     question_embedding = embed(  
148         question  
149     )  
150       
151     results = collection.query(  
152         query_embeddings=[  
153             question_embedding  
154         ],  
155         n_results=k  
156     )  
157       
158     return (  
159         results["documents"][0],  
160         results["metadatas"][0]  
161     )  
162   
163   
164   
165 def build_prompt(  
166     question: str,  
167     chunks: list[str]  
168 ) -> str:  
169       
170     context = "\n\n".join(  
171         chunks  
172     )  
173       
174     return f"""  
175     Answer the question using ONLY the context below.
```

```

rag_pipeline.py > ...
165     def build_prompt(
177         If the answer is not contained in the context,
178         say exactly:
179
180         I don't have that information.
181
182         Do not use outside knowledge.
183         Do not guess.
184         Do not invent facts.
185
186         Context:
187
188         {context}
189
190         Question:
191
192         {question}
193         """

```

```

rag_pipeline.py > ...
264     def ask_question(
305         answer = generate_answer(
306             question,
307             chunks
308         )

```

Output:

```

(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartC-Manual-RAG-Pipeline\rag-demo> .\.venv\Scripts\python.exe rag_pipeline.py
Added document: magic_school.txt
Added document: mars_mission.txt
Added document: ocean.txt
Added document: rainforest.txt

=====
Question:
How did Lena survive on Mars?

--- Retrieved Chunks ---

Chunk 1:
The Mars Horizon mission sends a team of astronauts to explore Mars.

During the mission, engineer Lena Ortiz becomes separated from the rest of the crew after a severe dust storm. She must use scientific knowledge and limited supplies to survive.

Lena grows food inside the habitat, repairs communi

Chunk 2:
ena grows food inside the habitat, repairs communication equipment, and sends a signal back to Earth while waiting for rescue.

Chunk 3:
ine biodiversity, and changes in sea temperature.

--- Final Answer ---
Lena survived by using her scientific knowledge and the limited supplies she had. She grew food inside the habitat, repaired the communication equipment, and sent a signal back to Earth while she waited for rescue.

```

3. Ask an Unrelated Question

Code:

```
rag_pipeline.py > ...
165     def build_prompt(
174         return f"""
175         Answer the question using ONLY the context below.
176
177         If the answer is not contained in the context,
178         say exactly:
179
180         I don't have that information.
181
182         Do not use outside knowledge.
183         Do not guess.
184         Do not invent facts.
```

```
rag_pipeline.py > ...
337     ask_question(
338         "Who won the football World Cup in 2022?"
339     )
```

Output:

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartC-Manual-RAG-Pipeline\rag-demo> .\.venv\Scripts\python.exe rag_pipeline.py
Added document: magic_school.txt
Added document: mars_mission.txt
Added document: ocean.txt
Added document: rainforest.txt

=====
Question:
Who won the football World Cup in 2022?

--- Retrieved Chunks ---

Chunk 1:

Deforestation threatens these ecosystems by destroying habitats and reducing biodiversity.

Chunk 2:
The Pacific Ocean is the largest and deepest ocean on Earth.

It covers a huge portion of the planet and contains deep trenches, coral reefs, islands, and many marine ecosystems.

Scientists study the Pacific Ocean to understand climate patterns, marine biodiversity, and changes in sea temperature.

Chunk 3:
student, Adrian, discovers that an old dark sorcerer is trying to return to power. Adrian and his friends use what they learned at the academy to protect the school.

--- Final Answer ---
I don't have that information.
```

4. Source Attribution

Code:

```

rag_pipeline.py > ...
264     def ask_question(
320         print(
321             "\n--- Sources ---"
322         )
323     )
324
325     for metadata in metadatas:
326
327         print(
328             f"Source: {metadata['source']} "
329             f"| Chunk: {metadata['chunk']}"
330         )

```

Output:

```

(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Wexa Solutions\Week 5\PartC-Manual-RAG-Pipeline\rag-demo> .\venv\Scripts\python.exe rag_pipeline.py
Added document: rainforest.txt

=====
Question:
How did Lena survive on Mars?

--- Retrieved Chunks ---

Chunk 1:
The Mars Horizon mission sends a team of astronauts to explore Mars.

During the mission, engineer Lena Ortiz becomes separated from the rest of the crew after a severe dust storm. She must use scientific knowledge and limited supplies to survive.

Lena grows food inside the habitat, repairs communi

Chunk 2:
ena grows food inside the habitat, repairs communication equipment, and sends a signal back to Earth while waiting for rescue.

Chunk 3:
ine biodiversity, and changes in sea temperature.

--- Final Answer ---
Lena survived by using her scientific knowledge and the limited supplies she had. She grew food inside the habitat, repaired the communication equipment, and sent a signal back to Earth while she waited for rescue.

--- Sources ---
Source: mars_mission.txt | Chunk: 0
Source: mars_mission.txt | Chunk: 1
Source: ocean.txt | Chunk: 1
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Wexa Solutions\Week 5\PartC-Manual-RAG-Pipeline\rag-demo>

```

PART D: RAG QUALITY & HALLUCINATION REDUCTION

Exercises:-

1. Build a Table of 5 Test Questions

Code:

rag_evaluation.py > ...

```
47 TEST_QUESTIONS = [  
48     {  
49         "question":  
50             "How did Lena survive on Mars?",  
51  
52         "expected":  
53             "She used scientific knowledge, "  
54             "grew food, repaired communication "  
55             "equipment, and sent a signal to Earth.",  
56  
57         "expected_source":  
58             "mars_mission.txt"  
59     },  
60  
61     {  
62         "question":  
63             "What subjects do students study "  
64             "at the Academy of Arcane Arts?",  
65  
66         "expected":  
67             "Spell casting, potion making, "  
68             "magical creatures, and defensive magic.",  
69  
70         "expected_source":  
71             "magic_school.txt"  
72     },  
73  
74     {  
75         "question":  
76             "What threatens biodiversity "  
77             "in the Amazon rainforest?",  
78  
79         "expected":  
80             "Deforestation destroys habitats "
```

rag_evaluation.py > ...

```
81         "and reduces biodiversity.",
82
83         "expected_source":
84             "rainforest.txt"
85     },
86
87     {
88         "question":
89             "What is special about the Pacific Ocean?",
90
91         "expected":
92             "It is the largest and deepest "
93             "ocean on Earth.",
94
95         "expected_source":
96             "ocean.txt"
97     },
98
99     {
100         "question":
101             "Where is the Academy of Arcane Arts located?",
102
103         "expected":
104             "Inside an ancient castle surrounded "
105             "by enchanted forests.",
106
107         "expected_source":
108             "magic_school.txt"
109     }
110 ]
```

Output:

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> .\.venv\Scripts\python.exe rag_evaluation.py

TEST QUESTION 1
Question: How did Lena survive on Mars?
Expected answer: She used scientific knowledge, grew food, repaired communication equipment, and sent a signal to Earth.
Expected source: mars_mission.txt
Retrieved sources: ['mars_mission.txt', 'mars_mission.txt', 'ocean.txt']

Generated answer:
Lena survived by using her scientific knowledge and the limited supplies she had. She grew food inside the habitat, repaired the communication equipment,
ent a signal back to Earth while she waited for rescue.

Manual evaluation:
Retrieval correct? YES / NO
Answer correct? YES / NO
Failure type if any: Retrieval / Generation / None
-----

TEST QUESTION 2
Question: What subjects do students study at the Academy of Arcane Arts?
Expected answer: Spell casting, potion making, magical creatures, and defensive magic.
Expected source: magic_school.txt
Retrieved sources: ['magic_school.txt', 'magic_school.txt', 'rainforest.txt']

Generated answer:
Students study spell casting, potion making, magical creatures, and defensive magic.

Manual evaluation:
Retrieval correct? YES / NO
Answer correct? YES / NO
```

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> .\.venv\Scripts\python.exe rag_evaluation.py

TEST QUESTION 3
Question: What threatens biodiversity in the Amazon rainforest?
Expected answer: Deforestation destroys habitats and reduces biodiversity.
Expected source: rainforest.txt
Retrieved sources: ['rainforest.txt', 'rainforest.txt', 'ocean.txt']

Generated answer:
Deforestation.

Manual evaluation:
Retrieval correct? YES / NO
Answer correct? YES / NO
Failure type if any: Retrieval / Generation / None
-----

TEST QUESTION 4
Question: What is special about the Pacific Ocean?
Expected answer: It is the largest and deepest ocean on Earth.
Expected source: ocean.txt
Retrieved sources: ['ocean.txt', 'ocean.txt', 'rainforest.txt']

Generated answer:
The Pacific Ocean is the largest and deepest ocean on Earth, covering a huge portion of the planet and containing deep trenches, coral reefs, islands, and many
marine ecosystems.
```

```
TEST QUESTION 5
Question: Where is the Academy of Arcane Arts located?
Expected answer: Inside an ancient castle surrounded by enchanted forests.
Expected source: magic_school.txt
Retrieved sources: ['magic_school.txt', 'magic_school.txt', 'mars_mission.txt']

Generated answer:
The Academy of Arcane Arts is located inside an ancient castle surrounded by enchanted forests.

Manual evaluation:
Retrieval correct? YES / NO
Answer correct? YES / NO
Failure type if any: Retrieval / Generation / None
-----
```

Table:

No.	Question	Expected Answer	Expected Source
1	How did Lena survive on Mars?	Used science, grew food, repaired communication equipment, sent a signal to Earth	mars_mission.txt
2	What subjects do students study at the Academy?	Spell casting, potion making, magical creatures, defensive magic	magic_school.txt

3	What threatens Amazon biodiversity?	Deforestation	rainforest.txt
4	What is special about the Pacific Ocean?	Largest and deepest ocean on Earth	ocean.txt
5	Where is the Academy located?	Ancient castle surrounded by enchanted forests	magic_school.txt

2. Evaluate Retrieval and Answer Quality

Code:

```
rag_evaluation.py > ...
312 def run_evaluation(
360     retrieved_sources = [
361         metadata["source"]
362         for metadata in metadatas
363     ]
364
365
366     print(
367         f"\nTEST QUESTION {number}"
368     )
369
370     print(
371         "Question:",
372         question
373     )
374
375     print(
376         "Expected answer:",
377         expected
378     )
379
380     print(
381         "Expected source:",
382         expected_source
383     )
384
385     print(
386         "Retrieved sources:",
387         retrieved_sources
388     )
389
390     print(
391         "\nGenerated answer:"
392     )
393
394     print(
395         answer
396     )
```

```

rag_evaluation.py > ...
312     def run_evaluation(
402         print(
403             "Retrieval correct? YES / NO"
404         )
405
406         print(
407             "Answer correct? YES / NO"
408         )

```

Output:

```

(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> .\venv\Scripts\python.exe rag_evaluation.py

=====
EVALUATION WITH CHUNK SIZE = 300
=====

TEST QUESTION 1
Question: How did Lena survive on Mars?
Expected answer: She used scientific knowledge, grew food, repaired communication equipment, and sent a signal to Earth.
Expected source: mars_mission.txt
Retrieved sources: ['mars_mission.txt', 'mars_mission.txt', 'ocean.txt']

Generated answer:
Lena survived by using her scientific knowledge and the limited supplies she had. She grew food inside the habitat, repaired the communication equipment, and sent a signal back to Earth while she waited for rescue.

```

```

(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> .\venv\Scripts\python.exe rag_evaluation.py

TEST QUESTION 2
Question: What subjects do students study at the Academy of Arcane Arts?
Expected answer: Spell casting, potion making, magical creatures, and defensive magic.
Expected source: magic_school.txt
Retrieved sources: ['magic_school.txt', 'magic_school.txt', 'rainforest.txt']

Generated answer:
Students study spell casting, potion making, magical creatures, and defensive magic.

```

```

(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> .\venv\Scripts\python.exe rag_evaluation.py

TEST QUESTION 3
Question: What threatens biodiversity in the Amazon rainforest?
Expected answer: Deforestation destroys habitats and reduces biodiversity.
Expected source: rainforest.txt
Retrieved sources: ['rainforest.txt', 'rainforest.txt', 'ocean.txt']

Generated answer:
Deforestation.

```

```

(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> .\venv\Scripts\python.exe rag_evaluation.py

TEST QUESTION 4
Question: What is special about the Pacific Ocean?
Expected answer: It is the largest and deepest ocean on Earth.
Expected source: ocean.txt
Retrieved sources: ['ocean.txt', 'ocean.txt', 'rainforest.txt']

Generated answer:
The Pacific Ocean is the largest and deepest ocean on Earth, covering a huge portion of the planet and containing deep trenches, coral reefs, islands, and many marine ecosystems.

```

```

(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> .\venv\Scripts\python.exe rag_evaluation.py

TEST QUESTION 5
Question: Where is the Academy of Arcane Arts located?
Expected answer: Inside an ancient castle surrounded by enchanted forests.
Expected source: magic_school.txt
Retrieved sources: ['magic_school.txt', 'magic_school.txt', 'mars_mission.txt']

Generated answer:
The Academy of Arcane Arts is located inside an ancient castle surrounded by enchanted forests.

```

Table:

Question	Retrieval Correct?	Answer Correct?
Q1	Yes	Yes
Q2	Yes	Yes
Q3	Yes	Yes
Q4	Yes	Yes
Q5	Yes	Yes

3. Diagnose Retrieval vs Generation Failures

Code:

```
rag_evaluation.py > ...
312 def run_evaluation(
410     print(
411         "Failure type if any: "
412         "Retrieval / Generation / None"
413     )
```

Output:

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> .\.venv\Scripts\python.exe rag_evaluation.py

=====
EVALUATION WITH CHUNK SIZE = 300
=====

TEST QUESTION 1
Question: How did Lena survive on Mars?
Expected answer: She used scientific knowledge, grew food, repaired communication equipment, and sent a signal to Earth.
Expected source: mars_mission.txt
Retrieved sources: ['mars_mission.txt', 'mars_mission.txt', 'ocean.txt']

Generated answer:
Lena survived by using her scientific knowledge and the limited supplies she had. She grew food inside the habitat, repaired the communication equipment, and sent a signal back to Earth while she waited for rescue.

(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> .\.venv\Scripts\python.exe rag_evaluation.py

TEST QUESTION 2
Question: What subjects do students study at the Academy of Arcane Arts?
Expected answer: Spell casting, potion making, magical creatures, and defensive magic.
Expected source: magic_school.txt
Retrieved sources: ['magic_school.txt', 'magic_school.txt', 'rainforest.txt']

Generated answer:
Students study spell casting, potion making, magical creatures, and defensive magic.

(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> .\.venv\Scripts\python.exe rag_evaluation.py

TEST QUESTION 3
Question: What threatens biodiversity in the Amazon rainforest?
Expected answer: Deforestation destroys habitats and reduces biodiversity.
Expected source: rainforest.txt
Retrieved sources: ['rainforest.txt', 'rainforest.txt', 'ocean.txt']

Generated answer:
Deforestation.
```

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> .\.venv\Scripts\python.exe rag_evaluation.py
TEST QUESTION 4
Question: What is special about the Pacific Ocean?
Expected answer: It is the largest and deepest ocean on Earth.
Expected source: ocean.txt
Retrieved sources: ['ocean.txt', 'ocean.txt', 'rainforest.txt']

Generated answer:
The Pacific Ocean is the largest and deepest ocean on Earth, covering a huge portion of the planet and containing deep trenches, coral reefs, islands, and many marine ecosystems.
```

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> .\.venv\Scripts\python.exe rag_evaluation.py
TEST QUESTION 5
Question: Where is the Academy of Arcane Arts located?
Expected answer: Inside an ancient castle surrounded by enchanted forests.
Expected source: magic_school.txt
Retrieved sources: ['magic_school.txt', 'magic_school.txt', 'mars_mission.txt']

Generated answer:
The Academy of Arcane Arts is located inside an ancient castle surrounded by enchanted forests.
```

Table:

Question	Failure?	Failure Type	Explanation
Q1	No	None	Expected and actual answer match
Q2	No	None	Expected and actual answer match
Q3	No	None	Expected and actual answer match
Q4	No	None	Expected and actual answer match
Q5	No	None	Expected and actual answer match

4. Halve the Chunk Size and Compare Results

Code:

```
rag_evaluation.py > ...
420 if __name__ == "__main__":
421
422     run_evaluation(
423         chunk_size=300
424     )
425
426     run_evaluation(
427         chunk_size=150
428     )
```

```
rag_evaluation.py > ...
133 def chunk_text(
134     text: str,
135     chunk_size: int,
136     overlap: int = 50
137 ) -> list[str]:
```

Output:

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Wexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> .\.venv\Scripts\python.exe rag_evaluation.py
=====
EVALUATION WITH CHUNK SIZE = 300
=====

TEST QUESTION 1
Question: How did Lena survive on Mars?
Expected answer: She used scientific knowledge, grew food, repaired communication equipment, and sent a signal to Earth.
Expected source: mars_mission.txt
Retrieved sources: ['mars_mission.txt', 'mars_mission.txt', 'ocean.txt']

Generated answer:
Lena survived by using her scientific knowledge and the limited supplies she had. She grew food inside the habitat, repaired the communication equipment, and sent a signal back to Earth while she waited for rescue.

Manual evaluation:
Retrieval correct? YES / NO
Answer correct? YES / NO
Failure type if any: Retrieval / Generation / None
-----

TEST QUESTION 2
Question: What subjects do students study at the Academy of Arcane Arts?
Expected answer: Spell casting, potion making, magical creatures, and defensive magic.
Expected source: magic_school.txt
Retrieved sources: ['magic_school.txt', 'magic_school.txt', 'rainforest.txt']

Generated answer:
Students study spell casting, potion making, magical creatures, and defensive magic.

Manual evaluation:
Retrieval correct? YES / NO
Answer correct? YES / NO
Failure type if any: Retrieval / Generation / None
-----

TEST QUESTION 3
Question: What threatens biodiversity in the Amazon rainforest?
Expected answer: Deforestation destroys habitats and reduces biodiversity.
Expected source: rainforest.txt
Retrieved sources: ['rainforest.txt', 'rainforest.txt', 'ocean.txt']
```

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Wexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> .\.venv\Scripts\python.exe rag_evaluation.py
Generated answer:
Deforestation.

Manual evaluation:
Retrieval correct? YES / NO
Answer correct? YES / NO
Failure type if any: Retrieval / Generation / None
-----

TEST QUESTION 4
Question: What is special about the Pacific Ocean?
Expected answer: It is the largest and deepest ocean on Earth.
Expected source: ocean.txt
Retrieved sources: ['ocean.txt', 'ocean.txt', 'rainforest.txt']

Generated answer:
The Pacific Ocean is the largest and deepest ocean on Earth, covering a huge portion of the planet and containing deep trenches, coral reefs, islands, and many marine ecosystems.

Manual evaluation:
Retrieval correct? YES / NO
Answer correct? YES / NO
Failure type if any: Retrieval / Generation / None
-----

TEST QUESTION 5
Question: Where is the Academy of Arcane Arts located?
Expected answer: Inside an ancient castle surrounded by enchanted forests.
Expected source: magic_school.txt
Retrieved sources: ['magic_school.txt', 'magic_school.txt', 'mars_mission.txt']

Generated answer:
The Academy of Arcane Arts is located inside an ancient castle surrounded by enchanted forests.

Manual evaluation:
Retrieval correct? YES / NO
Answer correct? YES / NO
Failure type if any: Retrieval / Generation / None
-----
```

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Wexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> .\.venv\Scripts\python.exe rag_evaluation.py
=====
EVALUATION WITH CHUNK SIZE = 150
=====

TEST QUESTION 1
Question: How did Lena survive on Mars?
Expected answer: She used scientific knowledge, grew food, repaired communication equipment, and sent a signal to Earth.
Expected source: mars_mission.txt
Retrieved sources: ['mars_mission.txt', 'mars_mission.txt', 'mars_mission.txt']

Generated answer:
Lena survived on Mars by growing food inside the habitat, repairing the communication equipment, and sending a signal back to Earth.

Manual evaluation:
Retrieval correct? YES / NO
Answer correct? YES / NO
Failure type if any: Retrieval / Generation / None
-----

TEST QUESTION 2
Question: What subjects do students study at the Academy of Arcane Arts?
Expected answer: Spell casting, potion making, magical creatures, and defensive magic.
Expected source: magic_school.txt
Retrieved sources: ['magic_school.txt', 'magic_school.txt', 'magic_school.txt']

Generated answer:
Students study spell casting, potion making, magical creatures, and defensive magic.

Manual evaluation:
Retrieval correct? YES / NO
Answer correct? YES / NO
Failure type if any: Retrieval / Generation / None
-----

TEST QUESTION 3
Question: What threatens biodiversity in the Amazon rainforest?
Expected answer: Deforestation destroys habitats and reduces biodiversity.
Expected source: rainforest.txt
Retrieved sources: ['rainforest.txt', 'rainforest.txt', 'rainforest.txt']
```

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Wexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> .\.venv\Scripts\python.exe rag_evaluation.py
Generated answer:
Deforestation threatens biodiversity in the Amazon rainforest.

Manual evaluation:
Retrieval correct? YES / NO
Answer correct? YES / NO
Failure type if any: Retrieval / Generation / None
-----

TEST QUESTION 4
Question: What is special about the Pacific Ocean?
Expected answer: It is the largest and deepest ocean on Earth.
Expected source: ocean.txt
Retrieved sources: ['ocean.txt', 'ocean.txt', 'ocean.txt']

Generated answer:
The Pacific Ocean is the largest and deepest ocean on Earth, covering a huge portion of the planet and containing deep trenches, coral reefs, islands, and ma
marine ecosystems.

Manual evaluation:
Retrieval correct? YES / NO
Answer correct? YES / NO
Failure type if any: Retrieval / Generation / None
-----

TEST QUESTION 5
Question: Where is the Academy of Arcane Arts located?
Expected answer: Inside an ancient castle surrounded by enchanted forests.
Expected source: magic_school.txt
Retrieved sources: ['magic_school.txt', 'magic_school.txt', 'magic_school.txt']

Generated answer:
The Academy of Arcane Arts is located inside an ancient castle surrounded by enchanted forests.

Manual evaluation:
Retrieval correct? YES / NO
Answer correct? YES / NO
Failure type if any: Retrieval / Generation / None
-----
> (.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Wexa Solutions\Week 5\PartD-RAG-Quality\rag-evaluation> 
```

Table:

Question	Chunk 300 Retrieval	Chunk 300 Answer	Chunk 150 Retrieval	Chunk 150 Answer	Result
Q1	Yes	Yes	Yes	Yes	Improved
Q2	Yes	Yes	Yes	Yes	Improved
Q3	Yes	Yes	Yes	Yes	Improved
Q4	Yes	Yes	Yes	Yes	Improved
Q5	Yes	Yes	Yes	Yes	Improved

PART E: ADDING A /ask ENDPOINT TO THE FASTAPI SERVICE

Exercises:-

1. Test /ask with an Answerable Question

Code:

```
main.py > ...
69 class AskRequest(BaseModel):
70     question: str
71
72
73 class AskResponse(BaseModel):
74     answer: str
75     sources: list[str]
```

```
main.py > ask
344 @app.post(
345     "/ask",
346     response_model=AskResponse
347 )
348 def ask(
349     req: AskRequest
350 ):
351
352     chunks, metadatas = retrieve(
353         req.question
354     )
```

```

main.py > ask
348 def ask(
367     prompt = build_prompt(
368         req.question,
369         chunks
370     )

```

```

main.py > ask
348 def ask(
425     sources = sorted(
426         set(
427             metadata["source"]
428             for metadata in metadatas
429         )
430     )
431
432
433     return AskResponse(
434         answer=answer,
435         sources=sources
436     )

```

Output:

The screenshot shows a web browser window with the address bar displaying '127.0.0.1:8000/docs'. The page title is 'Library AI Service' with version '1.0.0' and 'OAS 3.1' tags. Below the title, it says '/openapi.json' and 'Week 5 FastAPI service with manual RAG-powered question answering.'.

The 'default' section lists three endpoints:

- GET /health** Health
- POST /summarize** Summarize
- POST /ask** Ask

The 'Schemas' section lists four schemas:

- AskRequest** Expand all object
- AskResponse** Expand all object
- HTTPValidationError** Expand all object
- SummaryRequest** Expand all object

127.0.0.1:8000/docs#/default/ask_ask_post

POST /ask Ask

Parameters

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "question": "How did Lena survive on Mars?"
}
```

Execute Clear

Responses

127.0.0.1:8000/docs#/default/ask_ask_post

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/ask' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "question": "How did Lena survive on Mars?"
  }'
```

Request URL

http://127.0.0.1:8000/ask

Server response

Code Details

200

Response body

```
{
  "answer": "Lena survived by using her scientific knowledge and the limited supplies she had. She grew food inside the habitat, repaired the communication equipment, and sent a signal back to Earth while she waited for rescue.",
  "sources": [
    "mars_mission.txt",
    "ocean.txt"
  ]
}
```

Response headers

```
content-length: 270
content-type: application/json
date: Tue, 22 Sep 2026 14:24:06 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

2. Add Clean LLM Error Handling

Code:

```
main.py > ask
348 def ask(
373     try:
374
375         response = (
376             groq_client
377             .chat
378             .completions
379             .create(
380                 model=CHAT_MODEL,
381
382                 messages=[
383                     {
384                         "role": "user",
385                         "content": prompt
386                     }
387                 ],
388
389                 temperature=0,
390
391                 max_tokens=400
392             )
393         )
394
395
396         answer = (
397             response
398             .choices[0]
399             .message
400             .content
401         )
402
403
404         if not answer:
405
```

```

main.py > ask
348 def ask(
406     answer = (
407         "I don't have that information."
408     )
409
410
411 except Exception as error:
412
413     print(
414         f"LLM request failed: {error}"
415     )
416
417     raise HTTPException(
418         status_code=503,
419         detail=(
420             "The AI provider is currently unavailable."
421         )
422     )

```

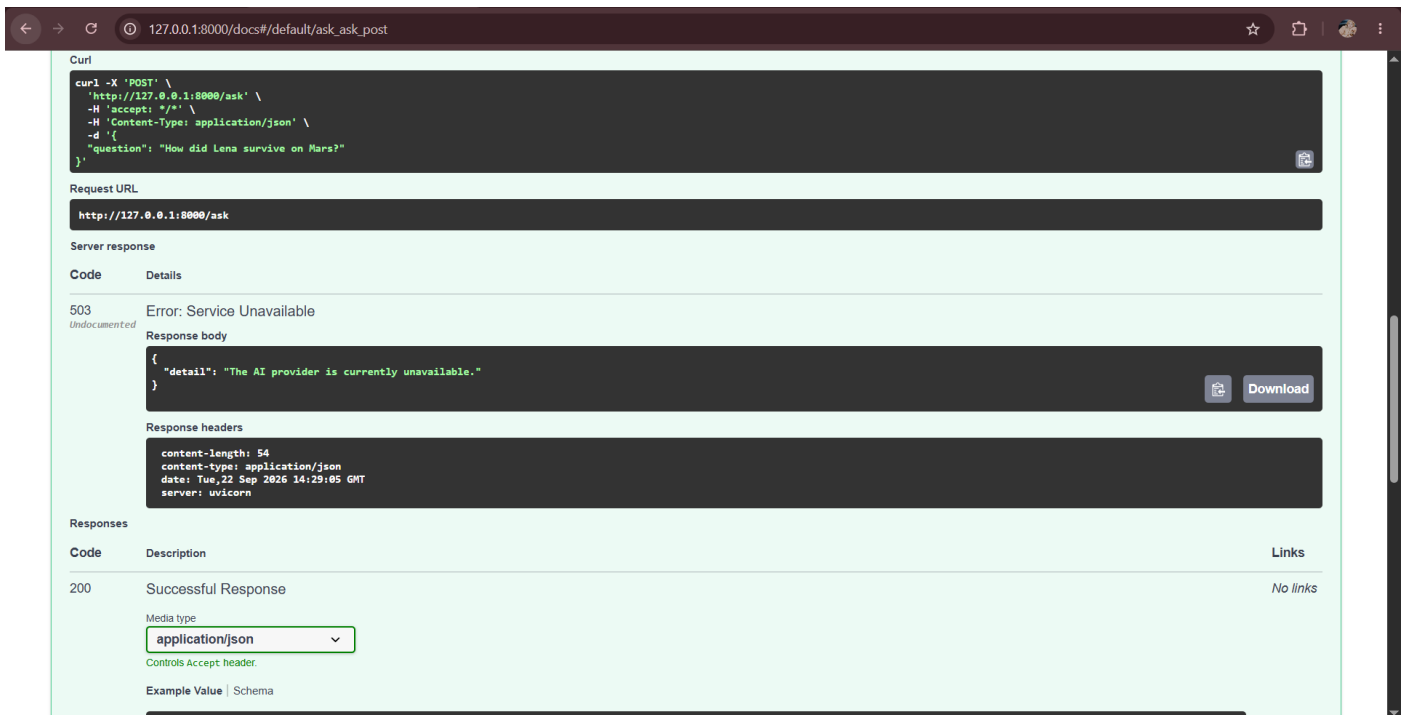
```

35
36 CHAT_MODEL = "invalid-model-for-test"
37

```

Output:

The screenshot shows a web browser window with a REST client interface. The URL bar shows the address `127.0.0.1:8000/docs#/default/ask_ask_post`. The interface is for a **POST** request to `/ask Ask`. It includes a **Parameters** section with "No parameters", a **Request body** section with a required field set to `application/json`, and a text area for the request body containing the JSON: `{ "question": "How did Lena survive on Mars?" }`. Below the text area are **Execute** and **Clear** buttons. At the bottom, there is a **Responses** section.



PART F: GIT: SAFELY UNDOING A COMMIT (git revert)

Exercises:-

1. Creating an Intentional Bad Commit

Code:

```
C: > Users > Eman > OneDrive > Desktop > Internship > Ne
55 def chunk_text(
56     text: str,
57     chunk_size: int = 0,
58     overlap: int = 50
59 ) -> list[str]:
```

Output:

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions> git commit -m "test: introduce invalid RAG chunk size"
[practice/git-revert 70f1250] test: introduce invalid RAG chunk size
1 file changed, 1 insertion(+), 1 deletion(-)
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions>
```

2. Reverting the Bad Commit and Verifying the Reverted File

Code:

```
C: > Users > Eman > OneDrive > Desktop > Internship > Ne
55 def chunk_text(
56     text: str,
57     chunk_size: int = 0,
58     overlap: int = 50
59 ) -> list[str]:
```

Output:

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions> git revert 70f1250
[practice/git-revert fae469e] Revert "test: introduce invalid RAG chunk size"
1 file changed, 1 insertion(+), 1 deletion(-)
```

Code:

```
C: > Users > Eman > OneDrive > Desktop > Internship > Nexa Solutions > Week 5 >

55     def chunk_text(
56         text: str,
57         chunk_size: int = 300,
58         overlap: int = 50
59     ) -> list[str]:
```

Output:

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions> git diff HEAD~1 HEAD
diff --git a/Week 5/PartC-Manual-RAG-Pipeline/rag-demo/rag_pipeline.py b/Week 5/PartC-Manual-RAG-Pipeline/rag-demo/rag_pipeline.py
index b940c7e..b826492 100644
--- a/Week 5/PartC-Manual-RAG-Pipeline/rag-demo/rag_pipeline.py
+++ b/Week 5/PartC-Manual-RAG-Pipeline/rag-demo/rag_pipeline.py
@@ -54,7 +54,7 @@ collection = chroma_client.create_collection(

     def chunk_text(
         text: str,
-        chunk_size: int = 0,
+        chunk_size: int = 300,
         overlap: int = 50
     ) -> list[str]:

(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions> █
```

3. Verifying Git History

Code:

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions> git log --oneline -6
```

Output:

```
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions> git log --oneline -6
fae469e (HEAD -> practice/git-revert) Revert "test: introduce invalid RAG chunk size"
70f1250 test: introduce invalid RAG chunk size
e5336ee (origin/main, origin/HEAD, main) Merge pull request #22 from eman-fatima46/feature/ai-ask-endpoint-practice
505b52b (origin/feature/ai-ask-endpoint-practice, feature/ai-ask-endpoint-practice) feat: add /ask endpoint wiring RAG pipeline into FastAPI service
fbc8f13 Merge pull request #21 from eman-fatima46/feature/rag-quality-evaluation
232b183 (origin/feature/rag-quality-evaluation, feature/rag-quality-evaluation) docs: add manual RAG evaluation results for 5 test questions
(.venv) PS C:\Users\Eman\OneDrive\Desktop\Internship\Nexa Solutions> █
```